

DDR and DRAM

Programs are first moved from SSD into DRAM by the operating system through paging. Once the data is inside DRAM, DDR prefetch refers to the internal DRAM mechanism where multiple adjacent data words are fetched from the memory array into an internal buffer and delivered to the memory controller as a burst. Prefetch is required because DRAM cells cannot run at I/O speeds, so multiple data words are fetched per array access and streamed out as a burst.

A RAM stick is physically a **DIMM (Dual Inline Memory Module)**. A DDR5 DIMM contains **multiple DRAM chips** mounted on the module, and the exact number of chips depends on the memory capacity and the data width of each DRAM device (for example, x8 or x16 devices).

x8 and x16 refer to the data width of an individual DRAM device, indicating how many data bits a single chip provides to the overall memory bus

In DDR5, the architectural organization changes compared to earlier DDR generations. **Each DDR5 DIMM is internally divided into two independent sub-channels**, commonly referred to as Channel A and Channel B. Each sub-channel is **32 bits wide**, allowing the memory controller to schedule commands and data transfers independently on each sub-channel, improving parallelism and efficiency.

Each sub-channel has:

- **32 data lines** ($DQ[31:0]$) for actual data transfer
- **Address lines** (around 21 bits, implementation-dependent) to specify rows, columns, banks, and bank groups
- **Control/command lines** (around 7 signals) used to encode memory commands such as ACTIVATE, READ, WRITE, and PRECHARGE

ACTIVATE (ACT) selects a **bank and a row** and opens that row by loading it into the sense amplifiers. This prepares the row so it can be accessed.

Once the row is open: **READ** selects a **column** from that open row and transfers the data out. **WRITE** selects a **column** from that open row and writes new data into the cells.

PRECHARGE (PRE) closes the open row by restoring data back into the cells, equalizing the bitlines to ~0.5 V, and returning the bank to an idle state so a **new ACT** can be issued.

Within a sub-channel, **all participating DRAM chips receive the same address and command signals simultaneously**. The data bus is split across multiple DRAM devices: for example, in a typical configuration using **x8 DRAM chips**, four chips together form one 32-bit sub-channel, with each chip contributing 8 data bits.

DDR5 also introduces **on-module power management**, where a dedicated **Power Management IC (PMIC)** is placed directly on the DIMM. This moves voltage regulation from the motherboard to the memory module, enabling better power control and supporting higher operating frequencies.

Overall, DDR5 improves performance not by widening the external data bus, but by **splitting each DIMM into independently operable sub-channels**, increasing concurrency, reducing command/data inefficiencies, and enabling better utilization of DRAM banks.

A DRAM chip consists of a **silicon die** housed inside a package. The package provides physical protection and signal routing, while the die contains the actual memory cells and supporting circuitry. Electrical connections to the outside world are made using a **Ball Grid Array (BGA)**, which carries data, address, command, power, and ground signals between the chip and the DIMM.

Internally, the die is organized into a hierarchical structure. In a typical **2 Gb DRAM chip**, the memory is divided into **8 bank groups**, each containing **4 banks**, giving a total of **32 banks**. This structure allows multiple memory operations to be overlapped and improves parallelism.

Each bank is arranged as a two-dimensional array of memory cells, commonly organized as **65,536 rows by 8,192 columns**. Each cell stores a single bit of data, and together these banks account for roughly **17 billion memory cells**, supported by a complex network of wordlines, bitlines, sense amplifiers, and peripheral control circuitry.

This internal organization enables DRAM to balance high density with high access bandwidth, forming the foundation for SDR, DDR, and LPDDR operation.

To uniquely access the approximately **17 billion memory cells** inside a DRAM chip, a **31-bit logical address** is required. This address is divided hierarchically to reflect the internal organization of the DRAM:

- **3 bits** select one of **8 bank groups**
- **2 bits** select one of **4 banks** within the chosen bank group
- **16 bits** select one of **65,536 rows** inside the bank
- **10 bits** select the column group, since **8,192 columns are accessed in groups of 8**

Although the full address is 31 bits, it is **not transmitted all at once**. To reduce pin count and improve efficiency, DRAM uses **time-multiplexed addressing** over approximately **21 address wires**. The address is sent in two phases:

1. **Bank group, bank, and row address** during the ACTIVATE command
2. **Column address** during the READ or WRITE command

For each access, multiple bits are transferred in parallel: a single operation reads or writes **8 bits simultaneously per device**, with wider data widths formed by multiple DRAM chips working together.

Each DRAM memory cell uses a **1T1C structure**, meaning **one transistor and one capacitor** are used to store **one bit of data**.

The **capacitor** physically stores the data as electric charge. It is typically implemented as a **deep trench or stacked structure in silicon**, only a few tens of nanometers in size. The capacitor consists of two conductive plates separated by a dielectric material. A **charged capacitor (~1V)** represents a binary **1**, while a **discharged capacitor (~0V)** represents a binary **0**.

The **access transistor** acts as a switch that controls when the capacitor can be read or written. Its gate is connected to the **wordline**, and its channel connects the capacitor to the **bitline**. When the wordline is asserted, the transistor turns on, allowing charge to flow between the capacitor and the bitline; when the wordline is deasserted, the transistor isolates the capacitor, preserving the stored value.

This simple 1T1C structure enables extremely high memory density but also requires periodic refreshing, since the stored charge naturally leaks over time.

In a **write operation**, the **wordline is asserted**, turning on the access transistor and connecting the capacitor to the bitline. Depending on the value being written, charge is either added to the capacitor to store a binary **1** or removed to store a binary **0**.

In a **read operation**, the wordline is again turned on so the capacitor connects to the bitline. The small amount of stored charge slightly changes the voltage on the bitline, and this change is sensed to determine whether the stored value is a **1** or a **0**.

When the wordline is turned **off**, the access transistor isolates the capacitor so it can retain its charge. However, because the capacitor and transistor are extremely small, **electrons slowly leak away over time**. As a result, DRAM cells cannot hold data indefinitely and must be **periodically refreshed** to restore the stored charge and maintain data integrity.

DRAM cells are arranged in a **two-dimensional memory array** formed by **wordlines** and **bitlines**.

- **Wordlines** run horizontally across the array and connect to the **gates of access transistors**.
- **Bitlines** run vertically and connect to the **transistor channels**, carrying data to and from the cells.

These lines are implemented in **different metal layers**, so they cross without making physical contact.

During operation, activating a **single wordline** connects **all cells in that row** to their corresponding bitlines simultaneously. At any given time, **only one wordline may be active** in a bank; activating multiple wordlines would cause charge sharing and data corruption.

Cell access is controlled hierarchically:

- A **row decoder**, driven by **16 address bits**, selects and activates exactly one wordline.
- A **column multiplexer**, driven by **10 address bits**, selects a group of **8 bitlines** to read or write data.

This organization allows DRAM to efficiently access large numbers of densely packed memory cells while maintaining correct and reliable operation.

During a **read operation**, the CPU issues a read command along with the target address. The appropriate **bank is selected**, and the **bitlines are precharged** to an intermediate voltage (typically around **0.5 V**). When the **wordline is activated**, all capacitors in the selected row connect to their respective bitlines. The tiny charge stored in each capacitor slightly raises or lowers the bitline voltage. **Sense amplifiers detect and amplify** this small voltage difference to full logic levels (**1 V or 0 V**). The **column multiplexer** then selects the required **8 bits**, which are transmitted back toward the CPU.

During a **write operation**, the CPU sends a write command, address, and the data to be stored. After bank selection and bitline precharge, the target **row is activated**, opening the cells to the bitlines. The **column multiplexer selects the target bitlines**, and **write drivers force the bitline voltages** to represent either a 1 or a 0. This voltage overwrites the charge in the capacitors, storing the new data.

Throughout both read and write operations, **multiple DRAM chips operate in parallel**, each contributing a portion of the total data width, enabling efficient and wide data transfers.

DRAM requires **periodic refresh** because the charge stored in each cell's capacitor **slowly leaks** through extremely small transistors, which would otherwise lead to data loss.

During a refresh operation, the **bitlines are precharged** to an intermediate voltage (around **0.5 V**), and **one row is opened at a time**. When the row is activated, the **sense amplifiers read the weakened charge and restore it** back to full logic levels, effectively rewriting the data into the cells.

This process is performed automatically by the memory system. A typical timing is **about 50 ns per row**, and **all rows in a bank must be refreshed within a 64 ms interval** to ensure data integrity.

A **row hit (page hit)** occurs when a memory access targets a column within a row that is already open in a bank. In this case, only the **column address (10 bits)** is needed, and the costly steps of closing and reopening a row are skipped. This results in **much faster access**, and multiple accesses can be served repeatedly from the same open row. Modern memory controllers actively optimize access patterns to **maximize row hits** for performance.

A **row miss** happens when the next access targets a **different row** in the same bank. The currently open row must first be **precharged (closed)**, and then the new row must be **activated (opened)**. This sequence makes row misses significantly slower than row hits. The latency of these operations is governed by key timing parameters:

- **CAS latency (CL)**: delay from column access to data output (row hit path)
- **RAS-to-CAS delay (tRCD)**: time required to open a row before a column access
- **Row precharge time (tRP)**: time to close an active row
- **Row active time (tRAS)**: minimum time a row must remain open

DRAM performance is improved by using **multiple banks**, which operate independently. While one bank is handling a row miss or refresh, other banks can serve accesses or keep rows open. This independence increases the **probability of row hits**, reduces average memory access time, and enables better parallelism.

Banks are further organized into **bank groups**. A typical structure consists of **8 bank groups with 4 banks each**, totaling **32 banks**. Bank groups help manage high-frequency operation and refresh activity. During refresh, **one bank per group can be refreshed while the remaining banks continue servicing requests**, minimizing performance impact.

To support fast sequential access, DRAM uses a **burst buffer optimization**. Each chip includes a **128-bit temporary buffer**. The column address is split into **6 bits** to select which set of cells is loaded into the buffer and **4 bits** to select data from within the buffer. This allows **128 cells to be loaded at once**, and **16 sequential accesses** to be performed efficiently, resulting in **1,024 bits accessed per chip per burst**. This design provides high-speed burst transfers while still allowing flexible random access.

Because a full DRAM bank is extremely large (on the order of **65,536 × 8,192 cells**), directly driving such long wordlines and bitlines would cause excessive delay and power consumption. To solve this, each bank is **hierarchically subdivided into smaller subarrays**, typically around **1,024 × 1,024 cells**. Each subarray has its own **local sense amplifiers** and participates in **hierarchical row decoding**, where global decoders select a subarray and local decoders activate the exact wordline. This subdivision shortens wordlines and bitlines, reduces capacitive load, allows smaller capacitors, enables faster transistor switching, and significantly improves overall performance.

DRAM sensing uses a **differential pair architecture**. Each sense amplifier is connected to **two complementary bitlines**, and adjacent rows alternate which bitline they connect to. At any time, only **one bitline is actively connected to a cell**, while the other remains passive. When sensing occurs, the small voltage difference is amplified so that one bitline is driven to logic high and the other to logic low, forming a **cross-coupled inverter pair** with opposite values.

This differential design provides several key advantages:

- **Simple and reliable precharging**, as both bitlines naturally equalize to about **0.5 V**
- **High noise immunity**, since information is stored as a voltage difference rather than an absolute value
- **Reduced parasitic capacitance**, because the passive bitline is not directly loaded by cells

Together, subarray hierarchy and differential sensing enable DRAM to scale to very large capacities while maintaining speed, signal integrity, and power efficiency.